

(20 percent $\div$ 10 percent = $2\times$) its invested capital, and the bad business earning 5 percent on invested capital is worth no more than half its invested capital (5 percent $\div$ 10 percent = $0.5\times$). For both the good and bad business the intrinsic value is “no more than” the value calculated. The quantitative value is the ceiling from which we must discount the valuations to account for the degree to which each business can sustain its return on capital and the risk of each business relative to the government bonds.

If the yield on the long-term taxable government bond changes, the intrinsic values of the businesses change too. If the yield on the government bond falls to 5 percent, the value of the bad business rises to no more than its invested capital (5 percent $\div$ 5 percent = $1\times$), and the value of the good business rises to no more than four times its invested capital (20 percent $\div$ 5 percent = $4\times$). On the other hand, if the yield on the government bond rises to say 20 percent, the value of the bad business falls to no more than one-quarter its invested capital (5 percent $\div$ 20 percent = $0.25\times$), and the value of the good business falls to no more than one times its invested capital (20 percent $\div$ 20 percent = $1\times$). “The rates of return that investors need from any kind of investment,” wrote Buffett in 1999, “are directly tied to the risk-free rate that they can earn from government securities.”³⁸

The basic proposition is this: What an investor should pay today for a dollar to be received tomorrow can only be determined by first looking at the risk-free interest rate.

Consequently, every time the risk-free rate moves by one basis point—by 0.01%—the value of every investment in the country changes. People can see this easily in the case of bonds, whose value is normally affected only by interest rates. In the case of equities or real estate or farms or whatever, other very important variables are almost always at work, and that means the effect of interest rate changes is usually obscured. Nonetheless, the effect—like the invisible pull of gravity—is constantly there.

While we initially made the simplifying assumption that all earnings are paid out, in actuality the proportion of earnings reinvested and the proportion paid out as dividends is hugely influential to the valuation. For the purposes of this exercise, let’s ignore the impact of taxes. With the long bond earning 10 percent, each dollar of earnings reinvested in the good business earning a 20 percent return on capital will immediately return up to 200 cents on the dollar in business value (20 percent $\div$ 10 percent), which is a very good return. Contrast this with the return to the owner of the bad business returning 5 percent on capital. Each dollar of earnings reinvested in that business will become 50 cents on the dollar in business value